DRIVEABILITY—DEDICATED COMPRESSED NATURAL GAS (CNG)—ROUGH IDLE OR NO START WHEN COLD—VEHICLES EQUIPPED WITH DEDICATED 4.6L OR 5.4L CNG ENGINE ONLY	Article No. 01-22-1

FORD: 1996-2001 CROWN VICTORIA
1997-2001 E SERIES, F-150

ISSUE

Some Dedicated Compressed Natural Gas (CNG) vehicles may exhibit a rough idle, or no start in cold conditions, but will restart after the vehicle is warmed.

ACTION

Clean the CNG fuel injectors using Isopropyl Alcohol and the cleaning procedure listed. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

NOTE
THIS PROCEDURE SHOULD ONLY BE PERFORMED BY A CERTIFIED CNG TECHNICIAN.

NOTE
FOLLOW ALL FUEL SYSTEM SAFETY PROCEDURES LISTED IN THE WORKSHOP MANUAL.

Perform normal Diagnostic Procedures per the appropriate Ford Workshop Manual. If it is identified that the injectors are partially or completely clogged (normally due to contaminated Fuel - Compressed Natural Gas), then perform the following procedure to clean the CNG Injectors:

1. Start vehicle and idle at 1500 rpm for 5 minutes and then turn engine off. If vehicle won't start due to Clogged Injectors, the injectors may be completely stuck closed. If this is the case, attempt to free them by applying 12 Volts directly to the injector coil:
   a. Apply battery voltage to the side of the injector connector that the red wire circuit 361 connects to.
   b. Ground the other side of the connector momentarily and listen for an audible click to confirm that the injector has opened.

NOTE
DO NOT GROUND INJECTOR FOR MORE THAN TEN SECONDS, PROLONGED EXPOSURE MAY RESULT IN DAMAGE TO THE INJECTOR.

   c. If an injector remains stuck, wait 10 seconds, and repeat Steps A and B.

NOTE
IF AN INJECTOR DOES NOT PRODUCE AN AUDIBLE CLICK AFTER THREE ATTEMPTS, REPLACE INJECTOR.

To replace Dedicated CNG Fuel Injectors: (E-Series must remove the engine access cover).

(1) Vent fuel system per section 310-00B of appropriate Workshop Manual.

(2) Remove all fuel rail supply line retaining bolts and disconnect electrical connectors on fuel rail supply line.

(3) Remove rail from injectors and set rail to the side.

(4) Replace injectors per Section 303-04 (Crown Victoria), 303-04D (F-150), 303-04G (E-Series).

(5) Reinstall all fuel rail supply line retaining bolts. Torque retainers to 8-10 N•m (71-88 lb-in.) and install electrical connectors on fuel rail supply line.

(6) Leak Check all connections removed for fuel injector replacement. Use a quality leak reactant or equivalent over fuel system pressure fittings and lines. Leaks from the system will exhibit soapy bubbles. A mild soap and water solution may be used as an alternative.

(7) Test drive vehicle to confirm correction of rough idle. IF condition is still present, continue with Step 2.

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2. Remove CNG Coalescent Filter Bowl after properly venting the Fuel Line (per Ford CNG Venting Procedure for Servicing Filter Bowl).
3. Check for dirt or oil in the filter bowl - clean if necessary.

NOTE
REPLACE COALESCENT FILTER ELEMENT IF IT APPEARS TO BE DISCOLORED WITH DARK DEPOSITS. COALESCENT FILTER ELEMENT PART NUMBER F5TZ-9365-AA.

4. Fill the filter bowl to the top with Isopropyl Alcohol (with strength of at least 90% by volume), with Coalescent filter in place.
5. Reinstall the bowl, torque bowl to: 40 ± 6 N•m (30 ± 5 lb-ft), Torque Drain Plug to: 10 ± 3 N•m (8 ± 2 lb-ft)
6. Start engine and run for about 5 minutes at 1500 rpm in Park.
7. Test Drive Vehicle to confirm correction of Rough Idle.

NOTE
IN SOME CASES, IF ROUGH IDLE REMAINS DUE TO TOUGH CLOGGING OF INJECTORS, REPEAT STEPS 2 THROUGH 7.

LABOR OPERATION CLAIMING CHART

Operation	Labor Description	Vehicle	Time
012201A	Apply voltage and perform fuel system cleaning procedure.	E-Series	2.2
		F-150	2.1
		Crown Vic	1.7
012201B	Apply voltage, perform fuel system cleaning procedure and replace one fuel injector.	E-Series	2.8
		F-150	2.5
		Crown Vic	2.0
012201C	Apply voltage, perform fuel system cleaning procedure and replace two or more fuel injectors.	E-Series	3.0
		F-150	2.7
		Crown Vic	2.1
012201D	Apply voltage, replace one fuel injector.	E-Series	1.9
		F-150	1.6
		Crown Vic	1.2
012201E	Apply voltage, replace two or more fuel injectors.	E-Series	2.0
		F-150	1.8
		Crown Vic	1.3

PART NUMBER	PART NAME
F5TZ-9365-AA	CNG - Coalescent Filter Element

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage And Emissions Warranty Coverage

DEALER CODING

BASIC PART NO.	CONDITION CODE
9F593	42

OASIS CODES: 601300, 602300, 603300, 607000, 607400, 607500, 607600, 607700, 608000, 608400, 617400, 618400, 619400